Non-generic ECDLP leads for prime, binary and Koblitz curves

Search round, 2026-10-18 · study `research/ecdlp_leads_20261018` · repository `aburan28/crypto`

Status. Proposals and derivations, plus one measured lead (L8: exact enumerations with run reports under `reports/`). No security claim about any deployed curve is made or withdrawn. Labels: *sourced* (cited fact), *derived* (follows by the argument shown; prose, no `IDC1` certificate yet), *computed* (shell arithmetic of a stated formula; data in `cover-genus-budget.csv`), *measured* (exact enumeration, run report), *proposed* (open). The Markdown `README.md` beside this PDF is the full text.

1. Question and scope

The owner asked for 10 to 33 new leads for *non-generic* ECDLP algorithms over prime-field, binary and Koblitz curves. A lead is admitted only if it passes three tests. It is not already on the repository's list. It names a mechanism that the Corrigan-Gibbs–Kogan bound for generic algorithms does not cover. It has a first experiment and a kill criterion. Ten leads passed. Ten ideas were closed on paper while drafting.

2. Bottom line

1. **Prime fields: one real lever, and it is curve-specific.** On curves with small integer coefficients the summation polynomial S_3 has small integer coefficients. Its small-root lattice collapses from 6 unknown monomials to 3. With projective (Farey) factor bases the 3 unknowns are also balanced in size. The level-1 reach for $m = 2$ rises from $p^{1/12}$ to $p^{1/9}$ to $p^{1/6}$ (derived). This applies to secp256k1, Curve25519, Curve448, BN254 and BLS12-381 G1. The NIST P-curves ($a = -3$) get $p^{1/10}$. Random-coefficient curves get nothing. Since $m = 2$ needs $1/2$ and $m = 4$ needs about $1/5$, this is an oracle improvement, not a threat.
2. **Koblitz curves: the cover route is not dead.** GHS is degenerate on Koblitz curves. Two facts reopen it. First, the Klein quartic over $\mathbb{F}_2$ has $L(X) = 1 + 5X^3 + 8X^6$, which equals $L\big(\mathrm{Res}_{\mathbb{F}_8/\mathbb{F}_2} E_{a=1}\big)$. Second, Eichler–Shimura applied to the CM newform of 49a1 twisted by a character χ of order n with $\chi(2) \neq 1$ gives a reduction isogenous to the trace-zero variety T_n, for every prime n. Class-group index calculus over $\mathbb{F}_2$ is subexponential. A covering curve of genus at most $c(n) \cdot n$ beats rho, where $c(131) \approx 1.9$ and $c(571) \approx 6.1$ (computed). The L8 étale search now covers every base curve of genus ≤ 3 over $\mathbb{F}_2$ (measured). T_n factors appear only at $n = 3$ and $n = 5$. The $n = 5$ case is one plane quartic with a genus-11 cover carrying T_5 for $a = 0$. A screen of every genus-4 curve plus a point-count lemma closes K-283 as well. The route is now closed for ECC2K-130, K-163, K-233 and K-283. A screen of every genus-5 curve (7,829 curves, 4,339 L-polynomials) finds no cover with $\lambda = \pm 1$ at $n = 409$ or 571 and excludes order 4 at $n = 409$; order 3 leaves 20 L-polynomials (40 curves) at K-409 and 12 (19 curves) at K-571 open.
3. **Random-b binary curves: only the cover question is new here.** A random-b curve does not lift to a Q-curve, so the Koblitz mechanisms do not transfer (derived sketch). LMFDB shows genus-5 Jacobians realising $\mathrm{Res}_{\mathbb{F}_{32}/\mathbb{F}_2}(E)$ for $t_5 \in \{-9, -7, -5, -3\}$, where GHS gives genus 16 (sourced). Whether an explicit map makes them usable is open (L10).

3. The ten leads

#	family	lead	status	first experiment	kill criterion
L8	Koblitz	Étale route: $D/\mathbb{F}_2$ with an order-n automorphism and an $\mathbb{F}_2$-map to E; its twist contains $E \times T_n$, genus $1 + n(g_0 - 1)$	measured ($g_0 \leq 3$; genus 4 screened): hits at $n = 3, 5$; closed for ECC2K-130, K-163, K-233, K-283	twisted L over $\mathbb{F}_8$ for the 59 genus-5 candidates with ord $\lambda = 3$ (K-409, K-571; genus 5 screened)	no genus-5 B passes the order-3 screen at $n = 409$
L6	Koblitz	Modular twists: $(f_E \otimes \chi) \bmod 2 \sim T_n$ for every n	derived	smallest curve containing $A_{f \otimes \chi}$ for $n = 3, 5, 7, 11$	genus grows faster than $c(n)n$
L10	binary	Genus-n Jacobians realising $\mathrm{Res}(E)$, beyond GHS	sourced at $n = 3, 5$ (LMFDB: genus-5 Jacobians for $t_5 \in \{-9, -7, -5, -3\}$)	explicit map $C \to E$ over $\mathbb{F}_{32}$ from the genus-5 census; then $n = 7$	no explicit map, or none at $n = 7$

L9	Koblitz and binary	Certified lower bounds on the cover genus	derived (partial); $(a = 0, n = 5)$ at genus 5 excluded (LMFDB)	Howe's test on T_n; point-count positivity	(produces bounds)
L1	prime	Coefficient collapse plus Farey bases	derived	Probes B/C with the collapsed and Farey lattices, $m = 2, 3, 4$, against random-coefficient controls	gain under $1.25\times$ at $m = 4$
L2	prime	Eisenstein–Farey bases for $j = 0$	derived	same harness, secp256k1-shaped toys	invariance lost under shifts
L7	Koblitz	Klein family: $\mathbb{Z}/n$ covers of E ramified at one place of degree $\mathrm{ord}_n(2)$	derived ($n = 3$)	$L(E, \chi, X)$ for $n = 7, 31, 73, 127$	no hit at 7 or 31
L3	prime	Algebraic-exposure ledger, 176 prime curves	proposed	table of super-monomial count × model × shape	none (audit)
L4	prime	Audit four unchecked prime-field IC papers	proposed	native re-implementation under matched rho	none below baseline
L5	prime	Global twist-collision relations	proposed	squarefree twist classes of $f(x)$, $x \le B$; ranks	yield below $B^{0.6}$

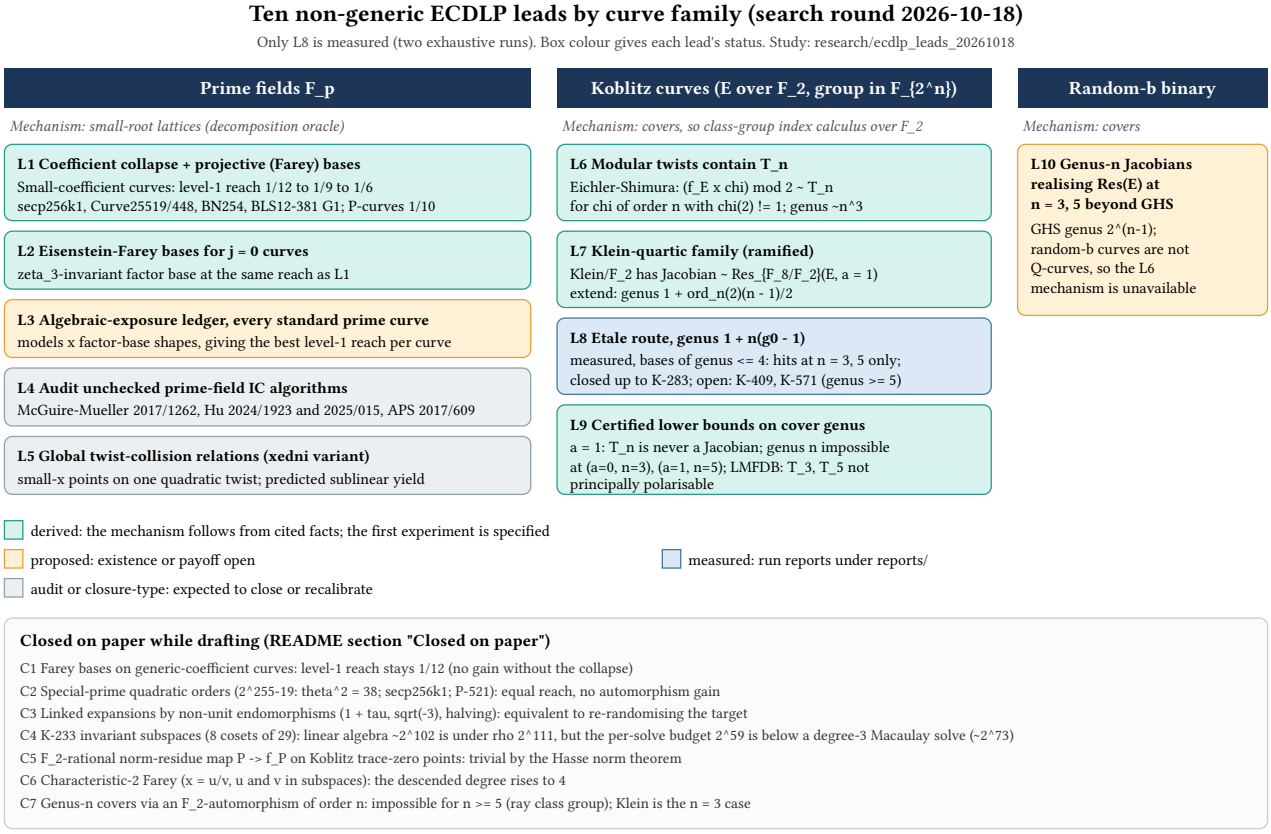


Figure 1: The ten leads by family with status colours, and the closures C1–C7. A diagram, not data.

4. Key derivations

L1, level-1 reach. $S_3(x_1, x_2, x_R) \equiv 0 \bmod p$ is a linear relation among unknown integer polynomials in the small summands. Reach is the largest factor base for which the product of their size bounds stays below p.

curve class	interval base	Farey base
generic a, b (5 unknowns plus a constant; Farey adds $s_1^2 s_2^2$)	1/12	1/12
$a = -3$, random b (P-256, P-384, P-521)	1/10	1/10
small a, b or small Montgomery A (3 unknowns c_2, c_1, c_0)	1/9	1/6

needed for decompositions to exist at $m = 2, 3, 4$	$1/2, 1/3, 1/4$	

For the collapsible class, $S_3 = c_2 x_R^2 - 2c_1 x_R + c_0$ with $c_2 = (x_1 - x_2)^2$, $c_1 = (x_1 + x_2)x_1 x_2 + 2b$ and $c_0 = (x_1 x_2)^2 - 4b(x_1 + x_2)$, all small integers. The Montgomery form is analogous with A. With $x = r/s$, each c_k becomes bihomogeneous of bidegree $(2, 2)$ and of size H^4. That gives $H^{12} \le p$ and $|F| \approx H^2 \le p^{1/6}$.

L6, Eichler–Shimura. $f_E \otimes \chi$ has level $49\ell^2$, character χ^2 and $a_2 = \chi(2)t$. At $p = 2$, $\mathrm{Frob}^2 - \chi(2)t\ \mathrm{Frob} + 2\chi(2)^2 = 0$, so $\mathrm{Frob} = \chi(2)\pi_E$. Over the conjugates these are exactly the eigenvalues of T_n. By Tate, $A_{f\otimes\chi} \bmod 2 \sim T_n$.

L7, Klein. The point counts are 3, 5, 24 over $\mathbb{F}_2, \mathbb{F}_4, \mathbb{F}_8$, so $L_{\text{Klein}} = 1 + 5X^3 + 8X^6$. For the $a = 1$ Koblitz curve $t_3 = -5$, so $L(\text{Res}) = 1 - t_3 X^3 + 8X^6$. These are equal, so by Tate the two varieties are isogenous.

L8, L9, closure C7. Tame $\mathbb{Z}/n$ ramification over $\mathbb{F}_2$ needs branch places of degree at least $\mathrm{ord}_n(2)$. Riemann–Hurwitz then leaves three options: quotient genus 0 or 1 (genus at least about $\mathrm{ord}_n(2)(n-1)/2$), or étale over a quotient of genus $g_0 \ge 2$ (genus $1 + n(g_0 - 1)$, needing $n \mid \#J_B(\mathbb{F}_2)$). Genus exactly n forces $n = 3$. If $J_C \sim T_n$ then $\#C(\mathbb{F}_{2^k}) = 2^k + 1 + t_k$ for $n \nmid k$. For $a = 1$ that gives 4 points over $\mathbb{F}_2$ but 2 over $\mathbb{F}_4$, which is impossible.

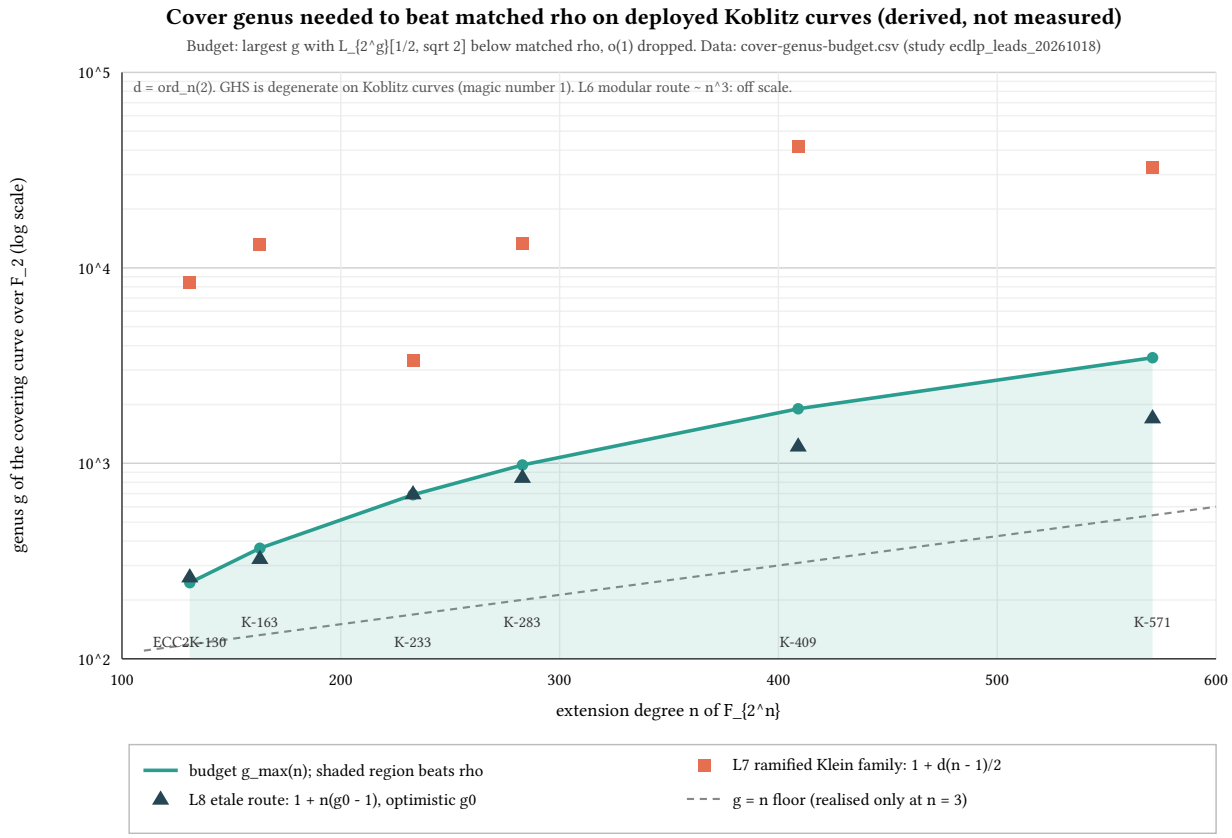


Figure 2: Derived, not measured. For each deployed Koblitz size: the L(1/2) genus budget $g_{\max}(n)$ (Enge–Gaudry constant $\sqrt{2}$, o(1) dropped), the étale-route genus $1 + n(g_0 - 1)$ at the smallest g_0 the Weil bound allows (optimistic), and the ramified Klein-family genus $1 + \mathrm{ord}_n(2)(n-1)/2$. Data: `cover-genus-budget.csv`.

5. L8 experiments (measured)

The native crate `etale_search` ran twice. Together the runs cover every curve $B/\mathbb{F}_2$ of genus 2 and 3:

- run-20261018T2043Z: genus 2 (768 models) and hyperelliptic genus 3 (6,144);
- run-20261018T2129Z: every smooth plane quartic (10,920 models, 78 isomorphism classes).

The model counts match the known masses of M_2, H_3 and $M_3 - H_3$ over $\mathbb{F}_2$ (sourced, Bergström 2008, 2009). Each $J_B(\mathbb{F}_2)$ was computed from relations. It was accepted with free rank 1 and torsion $= L_B(1)$, which proves the group because places of degree ≤ 3 generate Pic (derived). Every character χ of $J_B(\mathbb{F}_2)/n$, for every odd prime $n \mid \#J_B(\mathbb{F}_2)$, was tested for the factor $(1 - c\pi X)(1 - c\bar{\pi} X)$ in $L(B, \chi, X)$, in floating point and exactly in $\mathbb{Z}[\zeta_n]$. Every L-function passed the RH check and the functional equation. For quartics, 76,440 out-of-sample N_6 predictions passed. The exact and float tests agreed every time.

- **$n = 3$:** 192 characters in genus 2 and 576 in hyperelliptic genus 3, in six isogeny classes, for both $a = 0$ and $a = 1$. So T_3 sits in an explicit genus-4 Jacobian over $\mathbb{F}_2$ also for $a = 0$, where genus 3 is impossible.
- **$n = 5$:** 672 characters, all on one plane quartic with trivial automorphism group: $x^3y + x^2z^2 + xy^2z + xz^3 + y^3z = 0$, with $L_B = 1 + X + 2X^2 + 5X^3 + 4X^4 + 4X^5 + 8X^6$ and $J_B(\mathbb{F}_2) \cong \mathbb{Z}/25$. Its unramified $\mathbb{Z}/5$ covers of genus 11 carry T_5 of the $a = 0$ Koblitz curve in their Prym, up to isogeny.
- **Nothing else:** no hit for any other n, up to 71. In genus 2, one of the four L-polynomials the counting argument allows occurs ($1 + X + 2X^2 + 2X^3 + 4X^4$) and fails, so the genus-2 route is closed for $n \geq 5$.
- **Twisted covers (derived):** if Frobenius acts on $\mathbb{Z}/n$ by λ of order k, then T_n in the Prym needs $k \leq g_0 - 1$, and the cover exists only if $n \mid L_B(\lambda)$. The largest odd primes dividing $L(1)$ are 19, 37 and 71; for $L(-1)$ they are 19, 37 and 29 (genus 2, hyperelliptic genus 3, plane quartics).
- **Deployed sizes:**
  - ECC2K-130 ($g_0 \leq 2$) is closed by the Weil bound 34 (derived).
  - K-233 ($g_0 \leq 3$) is closed by the Weil bound 198 (derived).
  - K-163 ($g_0 \leq 3$) is closed because no genus ≤ 3 curve has $163 \mid L(\pm 1)$ (measured).
  - K-283 ($g_0 \leq 4$) is closed by the genus-4 screen (run-20261018T2144Z).
    - Every genus-4 curve over $\mathbb{F}_2$ has $\#J \leq 260$ and $|L(-1)| \leq 144$ (measured; the L-polynomial sets match Xarles's census).
    - The order-3 twist is excluded by a point-count lemma (derived). With $\operatorname{ord}\lambda = g_0 - 1$, $\#D(\mathbb{F}_{2^k}) = \#B(\mathbb{F}_{2^k}) + kt_k \equiv 0 \bmod n$, and for $a = 0$, $t_3 = 5 > 0$.
  - K-409 ($g_0 \leq 5$) and K-571 (≤ 7): genus 4 is closed for both. The genus-5 screen (run-20261103T2209Z) covers all 7,829 genus-5 curves over $\mathbb{F}_2$ and leaves both open at $g_0 = 5$.
    - Hyperelliptic and trigonal families enumerated independently (measured): 393,216 models give the stack count 512, 67,608 quintics singular at one point give 2,817, with the same L-sets as Dragutinović's lists. The 3,905 complete intersections are sourced: counts recounted and equal, smoothness not re-tested. The union has 4,339 distinct L-polynomials, as in the census.
    - No genus-5 curve has $n \mid L(\pm 1)$ for $n = 409, 571$. Order 4 at 409 is excluded by the lemma ($k = 4 = g_0 - 1$, $t_4 = 1 > 0$; derived).
    - Order 3 is not excluded (20 L-polynomials, 40 curves, at 409; 12, 19 curves, at 571): the Prym's spectrum is only forced to contain the μ_3-closure of T_n's, so the lemma does not apply.

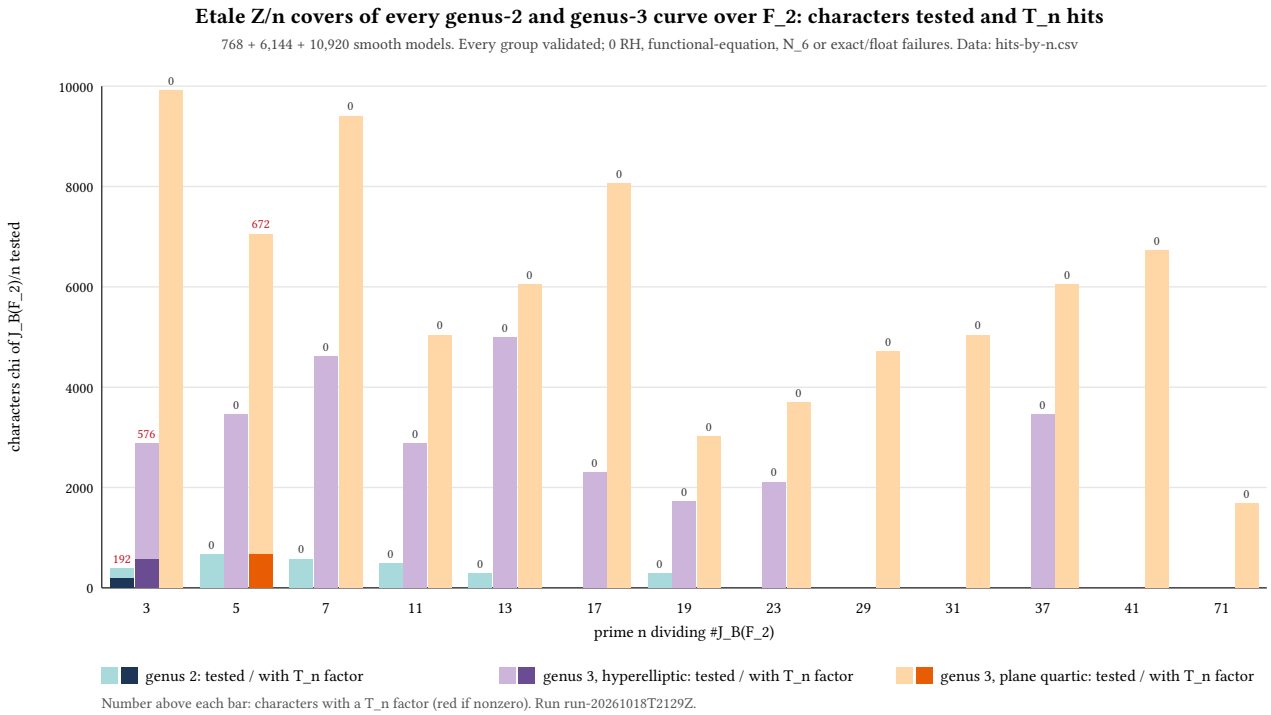


Figure 3: Run run-20261018T2129Z: characters tested and characters with a T_n factor, per n, over every genus-2 and genus-3 curve over $\mathbb{F}_2$. Recorded values; see the run report.

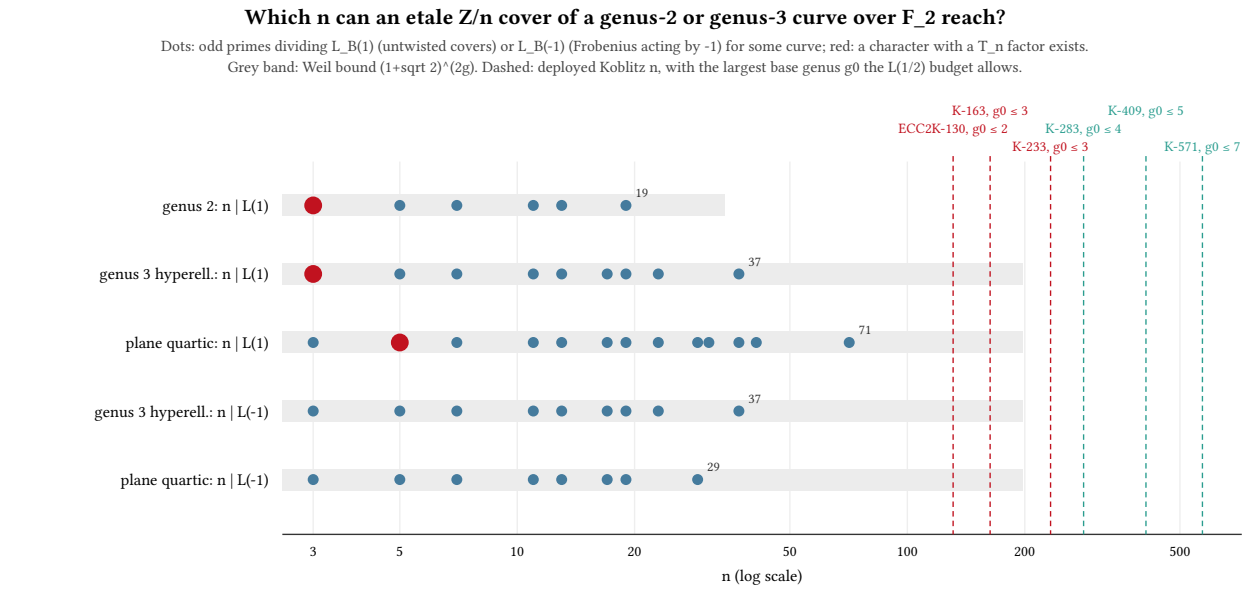


Figure 4: Run run-20261018T2129Z: odd primes dividing $L_B(\pm 1)$ over every genus-2 and genus-3 curve over $\mathbb{F}_2$, against the deployed Koblitz sizes and the base genus each budget allows. Recorded values plus the derived budget.

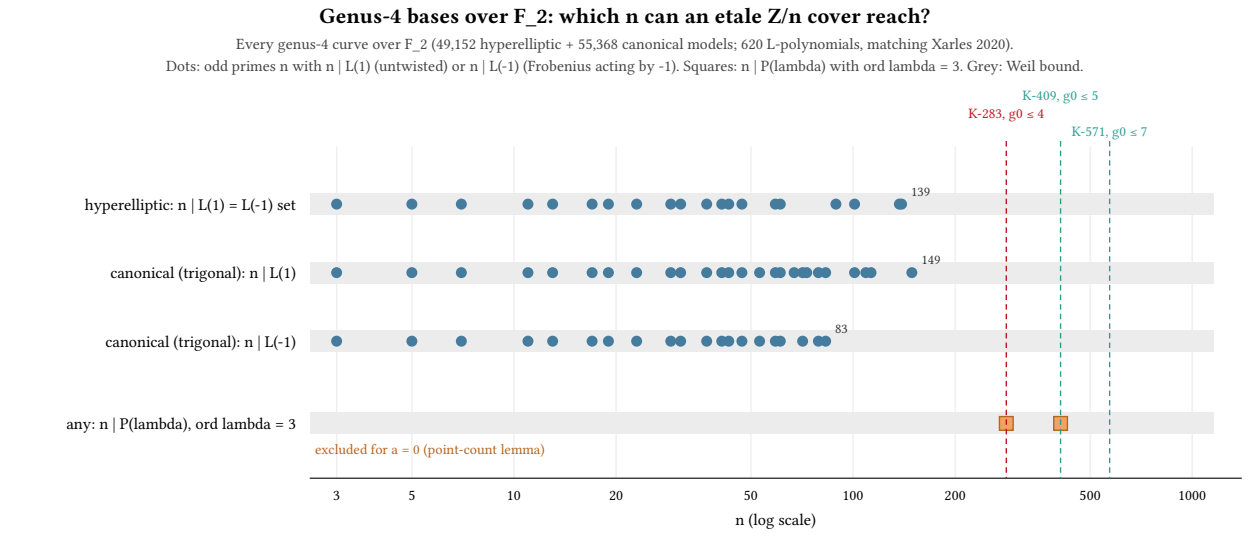
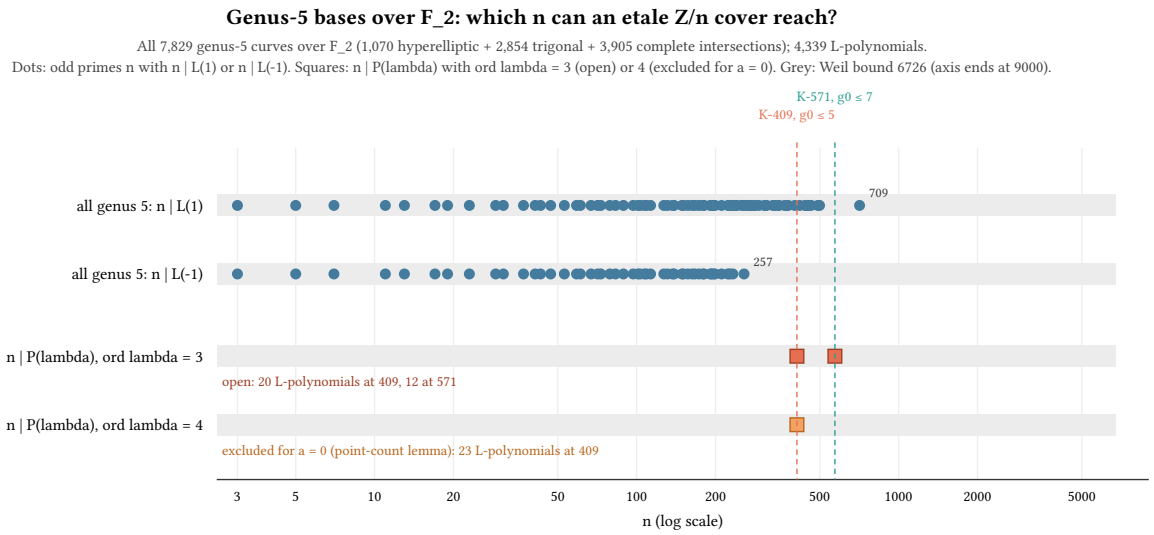


Figure 5: Run run-20261018T2144Z: odd primes dividing $L_B(\pm 1)$ over every genus-4 curve over $\mathbb{F}_2$, and the order-3 twist candidates, against the deployed sizes whose budget allows $g_0 = 4$. Recorded values plus the derived budget.



Orange dashed: g0 = 5 is the last base genus the budget allows (order-3 candidates remain). Teal dashed: g0 = 6, 7 also allowed, untested. Run run-20261103T2209Z.

Figure 6: Run run-20261103T2209Z: odd primes dividing $L_B(\pm 1)$ over every genus-5 curve over $\mathbb{F}_2$, with the order-3 candidates (open) and order-4 candidates (excluded), against the deployed sizes whose budget allows $g_0 \geq 5$. Recorded values plus the derived budget.

6. Closed on paper while drafting

C1	Farey bases on generic curves	the constant becomes a sixth unknown; reach stays $1/12$
C2	Special-prime quadratic orders (2^{255} − 19, secp256k1, P-521)	equal reach with the collapse, worse without it; no automorphism gain
C3	Linked expansions by $1+\tau$, $\sqrt{-3}$, halving	equivalent to re-randomising the target; towers raise the solving degree
C4	K-233 invariant subspaces (8 cosets of 29)	linear algebra $\approx 2^{102}$ is under rho $2^{110.9}$, but the per-solve budget 2^{59} is below a degree-3 Macaulay solve ($\approx 2^{73}$)
C5	K-571 and B-571 with $m = 5$ chained systems	needs solving degree at most 6 to 9 (the Semaev 2015 regime); the measured D growth contradicts it
C6	Oakley groups 3 and 4 by direct decomposition	per-trial Gröbner cost dominates; GHS stays the better route
C7	Genus-n covers with an $\mathbb{F}_2$-automorphism of order n	only $n = 3$ (Klein)
C8	Norm-residue map $P \mapsto f_P$ on Koblitz trace-zero points	trivial by the Hasse norm theorem
C9	Characteristic-2 Farey bases	the descended degree rises to 4
C10	Characters, Legendre symbols, pairings modulo small ℓ	these see only cofactors

7. Unresolved checks

- `identity.certificate/v1` records are owed for four claims: the L1/L2 counts, the Klein L-polynomial identity, the Eichler–Shimura twist, and the $a = 1$ obstruction.
- LMFDB returned fourteen classes.
  - T_3 and T_5, for $a = 0$ and $a = 1$, are simple, not principally polarisable, and contain no Jacobian.
  - At $n = 3$, $\mathrm{Res}_{\mathbb{F}_8/\mathbb{F}_2}(E)$ contains a genus-3 Jacobian exactly for $t_3 \in \{-5, -3, -1, 1\}$, the traces the point counts allow. The class for $t_3 = -5$ ($a = 1$ Koblitz) contains one Jacobian, the Klein quartic.
  - At $n = 5$, $\mathrm{Res}_{\mathbb{F}_{32}/\mathbb{F}_2}(E)$ contains genus-5 Jacobians for $t_5 \in \{-9, -7, -5, -3\}$ (1, 1, 1 and 4 of them), so L10′s realisations exist at $n = 5$, where GHS gives genus 16. It contains none for $t_5 = -11$, the $a = 0$ Koblitz curve: ($a = 0, n = 5$) is not realisable at genus 5. The open step for L10 is an explicit map $C \to E$ (Dragutinović's genus-5 census).

- Still blocked by a captcha: $t_5 \in \{-1, 1, 3, 5\}$, the two $n = 7$ trace-zero classes, and $t_3 = 5$ (already excluded by point counts).
- L8 covers bases of genus ≤ 3, and genus 4 and 5 by the screen. Not yet run: the twisted L-function over $\mathbb{F}_8$ for the 59 genus-5 order-3 candidates (K-409, K-571), and genus 6 and 7 bases for K-571; small-n hits of twisted ($\lambda = -1$) genus-3 covers; ramified and non-cyclic covers. The $n = 5$ genus-11 cover has no equations or certificate yet.
- The novelty audit against the cover-attack literature (Diem, Scholten, Hess, Ekedahl–Serre, Elkies) is owed before any of L6–L10 is called new.
- The L(1/2) budget is optimistic: it uses the hyperelliptic constant and drops o(1). General curves need slower class-group algorithms.
- Graphs checked and left unchanged: the IC scoreboard, the progress timeline, the IC leaderboard and the ECDLP attack matrix. The measured items, the two L8 enumerations, are lead-level searches with no cost or index-calculus result, and nothing here is certified.